

# Assignment 2: Process Mining and Automation

CSE346 Business Process Modeling

Spring 2026

## 1 Scope and Tool Choice

The assignment requires two deliverables: mine the provided event log with the Alpha algorithm, then automate the same steps using a tool or programming language. The selected automation language is Python because it handles CSV import, timestamp parsing, relation discovery, and reusable output generation without manual spreadsheet formulas.

**Online tools to use with the generated files:**

- Mermaid Live Editor for the generated PERT chart file `output/assignment2_pert.mmd`.
- diagrams.net if a polished submission figure is needed from the same dependency graph.
- ProM as an optional academic validation tool, since the process mining lecture introduces ProM and the Alpha algorithm.

The automation script is `tools/process_mining_alpha.py`. It reads `data/event_log.csv` and writes sorted logs, Alpha evidence tables, Mermaid PERT, DOT graph, and a text summary.

## 2 Lecture Concepts Applied

The process mining lecture defines the minimum event-log fields as case IDs and task IDs, with optional timestamp, resource, and data fields. This assignment uses all available fields: case ID, timestamp, activity, and resource. The same lecture defines the Alpha relations:

- Direct succession:  $x > y$  when  $x$  is directly followed by  $y$  in at least one case.
- Causality:  $x \rightarrow y$  when  $x > y$  and not  $y > x$ .
- Parallelism:  $x \parallel y$  when both  $x > y$  and  $y > x$ .
- Choice:  $x \# y$  when neither  $x > y$  nor  $y > x$ .

The process analysis lecture is used for interpretation: cycle time is the difference between case start and case end; waiting periods and handoffs indicate bottlenecks and improvement opportunities.

## 3 Preprocessing

The log was converted into a structured CSV with one row per event. Events were sorted by timestamp inside each case. One naming inconsistency in Case 3 was handled by trusting the activity label instead of the event ID prefix: the row whose ID begins with `Sh` is labeled “Get shipping address”, and the row whose ID begins with `Ge` is labeled “Ship product”. This preserves the activity sequence provided by the assignment text.

## 4 Generated Alpha-Algorithm Evidence

### 4.1 Traces

| Case | Trace                                                                                                                                                                            |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Check stock → Retrieve item → Confirm order → Get address → Emit invoice → Receive payment → Ship product → Archive order                                                        |
| 2    | Check stock → Check materials → Request materials → Obtain materials → Manufacture → Confirm order → Emit invoice → Get address → Ship product → Receive payment → Archive order |
| 3    | Check stock → Check materials → Manufacture → Confirm order → Emit invoice → Get address → Ship product → Receive payment → Archive order                                        |
| 4    | Check stock → Retrieve item → Confirm order → Get address → Emit invoice → Receive payment → Ship product → Archive order                                                        |

### 4.2 Direct Succession and Causality

| From              | To                | Count | Alpha relation |
|-------------------|-------------------|-------|----------------|
| Check materials   | Manufacture       | 1     | →              |
| Check materials   | Request materials | 1     | →              |
| Check stock       | Check materials   | 2     | →              |
| Check stock       | Retrieve item     | 2     | →              |
| Confirm order     | Emit invoice      | 2     | →              |
| Confirm order     | Get address       | 2     | →              |
| Emit invoice      | Get address       | 2     |                |
| Emit invoice      | Receive payment   | 2     | →              |
| Get address       | Emit invoice      | 2     |                |
| Get address       | Ship product      | 2     | →              |
| Manufacture       | Confirm order     | 2     | →              |
| Obtain materials  | Manufacture       | 1     | →              |
| Receive payment   | Archive order     | 2     | →              |
| Receive payment   | Ship product      | 2     |                |
| Request materials | Obtain materials  | 1     | →              |
| Retrieve item     | Confirm order     | 2     | →              |
| Ship product      | Archive order     | 2     | →              |
| Ship product      | Receive payment   | 2     |                |

### 4.3 Alpha Sets

- $T_W$  activities: Archive order, Check materials, Check stock, Confirm order, Emit invoice, Get address, Manufacture, Obtain materials, Receive payment, Request materials, Retrieve item, Ship product
- $T_I$  start activities: Check stock
- $T_O$  end activities: Archive order
- Parallel pairs: (Emit invoice, Get address), (Receive payment, Ship product)
- Main exclusive choices: available stock path versus manufacturing path; raw materials already available versus raw materials requested and obtained.

#### 4.4 Maximal Alpha Places

| Input activity set $A$                | Output activity set $B$             |
|---------------------------------------|-------------------------------------|
| $\{Checkmaterials\}$                  | $\{Manufacture, Requestmaterials\}$ |
| $\{Checkmaterials, Obtainmaterials\}$ | $\{Manufacture\}$                   |
| $\{Checkstock\}$                      | $\{Checkmaterials, Retrieveitem\}$  |
| $\{Confirmorder\}$                    | $\{Emitinvoice\}$                   |
| $\{Confirmorder\}$                    | $\{Getaddress\}$                    |
| $\{Emitinvoice\}$                     | $\{Receivepayment\}$                |
| $\{Getaddress\}$                      | $\{Shipproduct\}$                   |
| $\{Manufacture, Retrieveitem\}$       | $\{Confirmorder\}$                  |
| $\{Receivepayment\}$                  | $\{Archiveorder\}$                  |
| $\{Requestmaterials\}$                | $\{Obtainmaterials\}$               |
| $\{Shipproduct\}$                     | $\{Archiveorder\}$                  |

#### 4.5 Timing Metrics

| Case | Start            | End              | Cycle time |
|------|------------------|------------------|------------|
| 1    | 2012-07-30 11:14 | 2012-08-06 09:12 | 6.92 days  |
| 2    | 2012-08-01 09:44 | 2012-08-07 08:50 | 5.96 days  |
| 3    | 2012-08-02 10:06 | 2012-08-06 10:25 | 4.01 days  |
| 4    | 2012-08-04 08:11 | 2012-08-08 14:08 | 4.25 days  |

| From              | To                | Occurrences | Average hours |
|-------------------|-------------------|-------------|---------------|
| Ship product      | Receive payment   | 2           | 76.28         |
| Emit invoice      | Receive payment   | 2           | 57.32         |
| Get address       | Emit invoice      | 2           | 46.76         |
| Request materials | Obtain materials  | 1           | 44.63         |
| Obtain materials  | Manufacture       | 1           | 30.88         |
| Receive payment   | Ship product      | 2           | 14.42         |
| Ship product      | Archive order     | 2           | 11.22         |
| Check materials   | Manufacture       | 1           | 5.88          |
| Check stock       | Retrieve item     | 2           | 3.46          |
| Check materials   | Request materials | 1           | 1.10          |
| Get address       | Ship product      | 2           | 1.07          |
| Retrieve item     | Confirm order     | 2           | 0.47          |
| Confirm order     | Emit invoice      | 2           | 0.42          |
| Confirm order     | Get address       | 2           | 0.32          |
| Check stock       | Check materials   | 2           | 0.22          |
| Emit invoice      | Get address       | 2           | 0.20          |
| Manufacture       | Confirm order     | 2           | 0.13          |
| Receive payment   | Archive order     | 2           | 0.12          |

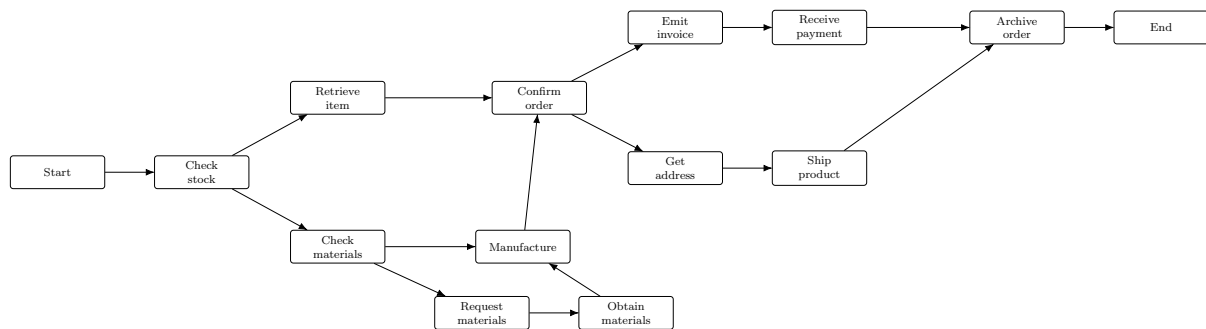
### 5 Mined Process Interpretation

The mined process starts with *Check stock availability* and ends with *Archive order*. The main behavior is:

- After stock check, there is an exclusive choice between retrieving an available warehouse item and entering the manufacturing/materials branch.
- In the manufacturing branch, checking materials may lead directly to manufacture or to requesting and obtaining raw materials first.
- Both warehouse retrieval and manufacturing converge at confirming the order.
- After confirmation, invoice/address and payment/shipping appear in different orders across traces. The Alpha algorithm therefore detects *Emit invoice* parallel with *Get shipping address*, and *Receive payment* parallel with *Ship product*.
- Archiving occurs after payment and shipment are complete.

## 6 PERT Chart

The following PERT-style network is the human-readable translation of the mined dependencies. The generated Mermaid version is available in `output/assignment2_pert.mmd`.



## 7 Automation Design

The Python script implements the following reusable workflow:

1. Read any CSV containing `case_id`, `timestamp`, `activity`, and `resource`.
2. Sort events per case by timestamp.
3. Extract traces, start activities  $T_I$ , end activities  $T_O$ , and task set  $T_W$ .
4. Count direct succession pairs.
5. Derive causality, parallel, and choice relations using the Alpha definitions from the lecture.
6. Build maximal Alpha places  $(A, B)$  where every activity in  $A$  causes every activity in  $B$ , activities inside  $A$  are pairwise choice, and activities inside  $B$  are pairwise choice.
7. Generate outputs for documentation and online tools.

Run command:

```
python3 tools/process_mining_alpha.py --input data/event_log.csv --output-dir output
```

Generated files:

| File | Purpose |
|------|---------|
|------|---------|

|                                               |                                                              |
|-----------------------------------------------|--------------------------------------------------------------|
| output/event_log_sorted.csv                   | Clean, sorted event log for checking preprocessing.          |
| output/assignment2_generated_tables_alpha.txt | Alpha evidence tables included in this report.               |
| output/assignment2_pert.mmd                   | Mermaid PERT chart for Mermaid Live Editor.                  |
| output/assignment2_alpha_net.dot              | DOT dependency graph for Graphviz or diagrams.net workflows. |
| output/assignment2_summary.txt                | Plain-text summary of counts, relations, and places.         |

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## 8 Findings and Improvement Opportunities

The average observed cycle time is about 5.28 days. The longest waiting periods occur around shipping/payment and invoice/payment timing, which suggests that the process has external waiting time and asynchronous handoffs. From the process analysis lecture, these delays affect cycle time and work-in-progress even when individual system tasks are short.

Recommended improvements are:

- Integrate payment and shipment status so the order can be archived only after both are complete and visible.
- Use automatic alerts for long gaps after invoice emission or shipment dispatch.
- Standardize whether payment should happen before shipment or whether both can proceed independently; the log shows both orders.
- Monitor resource handoffs between SYS1, SYS2, warehouse employees, and DMS because handoffs are common sources of delay and errors.

## 9 Conclusion

The Alpha algorithm discovers an order-fulfillment process with one start, one end, stock/manufacturing choices, and two parallel sections. The automated Python tool produces the same evidence required in Task One and provides reusable outputs for Task Two. The online visualization path is Mermaid Live Editor or diagrams.net for the PERT graph, with ProM available for optional validation against a process mining tool introduced in the lectures.